

Research Interests

My research interests lie in **Computer Vision, Multimodal Learning, Vision-Language Models (VLMs), and Trustworthy AI**. I am interested in developing **reliable, interpretable, and human-centered AI systems** that bridge visual perception and multimodal reasoning for real-world applications. My current research focuses on:

- (1) **Efficient visual understanding**, including object detection, scene understanding, and lightweight architectures for real-time inference;
- (2) **Multimodal learning**, integrating visual, linguistic, and contextual information for video understanding, video-to-audio generation, and vision-language reasoning;
- (3) **Human-centered AI applications**, leveraging multimodal perception to analyze complex real-world environments and support intelligent decision-making;
- (4) **Trustworthy AI**, with emphasis on robustness, interpretability, privacy-preserving machine learning, and the safe deployment of vision-language models.

Education

Sogang University (Transferred 2024)

Seoul, South Korea

B.S. IN COMPUTER SCIENCE

Expected Aug. 2026

- Relevant Coursework: Deep Learning, Machine Learning, Big Data Computing

Hankuk University of Foreign Studies (HUFS)

Seoul, South Korea

INFORMATION & COMMUNICATION ENGINEERING

2020 – 2023

- Transferred to Sogang University in 2024

Projects

MULTI-TRACK VIDEO-TO-AUDIO DESCRIPTION (4 MEMBERS)

Mar 2026 – Jun 2026

- Developed an end-to-end **Vision-Language Model (VLM)** pipeline that analyzes videos, generates structured multi-track audio (dialogue, sound effects, music, and ambience), and synthesizes the final mixed video.
- Designed a reference-based evaluation framework for improving semantic consistency between visual understanding and generated audio descriptions.
- **Gold Prize (1st Place)**, Sogang University Capstone Design Competition (2026).
- **Paper accepted to KCC 2026** (Multi-Track Video-to-Audio Description).
GitHub: github.com/sogang-v2a/v2a_inspect

CHEST X-RAY MULTI-DISEASE DETECTION SYSTEM (CXR-CAD) (5 MEMBERS, TEAM LEADER)

Mar 2026 – Jun 2026

- Developed a multi-label chest X-ray disease diagnosis system using deep learning with **Focal Loss** and **Grad-CAM**.
- Designed a deployment-oriented evaluation framework considering robustness, calibration, and distribution shifts for trustworthy clinical AI.
- **Encouragement Prize**, Sogang University Capstone Design Competition (2026)
- **Paper accepted to KCC 2026**.
GitHub: github.com/sogang-cxr-cad/CXR-CAD

RAG-BASED AI CRM MANAGER (5 MEMBERS, TEAM LEADER)

Sep 2025 – Dec 2025

- Built a **Multimodal RAG** system on GCP using GPT-4o, ChromaDB, FastAPI, and PostgreSQL for customer support automation.
- Developed autonomous customer-service and BI reporting agents through a full-stack architecture.
GitHub: github.com/suhwantcha/CS-Agent

FLIGHT DELAY PREDICTION (4 MEMBERS)

Nov 2025

- Built a flight arrival delay prediction model using an **XGBoost + LightGBM** ensemble with statistical feature engineering.
- **Excellence Award**, 2025 Chung-Ang × Sogang University Joint Datathon.
GitHub: github.com/suhwantcha/Flight-Delay-Prediction

PINTOS OPERATING SYSTEM

Sep 2025 – Dec 2025

- Implemented user programs, priority scheduling, file systems, and synchronization mechanisms in the Pintos educational operating system using C.
GitHub: github.com/suhwantcha/OS-Pintos-Project

YOLO-PSYCHO-ANALYSIS

Apr 2025 – May 2025

- Developed a YOLOv8-based psychological assessment system that analyzes hand-drawn house sketches based on HTP theory.
GitHub: github.com/suhwantcha/YOLO-Psycho-Analysis

Experience

Teaching Assistant, Basic Artificial Intelligence Programming SOGANG UNIVERSITY Assisted Python programming labs, answered student questions, graded assignments, and managed course administration.	Seoul, South Korea Mar 2026 – Jun 2026
Teaching Assistant, Understanding Artificial Intelligence SOGANG UNIVERSITY - Supported an introductory AI course for 50+ students through Q&A sessions, grading, and course management. - Conducted PyTorch-based laboratory sessions covering CNNs and Transformer models.	Seoul, South Korea Sep 2025 – Dec 2025
Teaching Assistant, Computer Architecture SOGANG UNIVERSITY Assisted students with computer architecture coursework and laboratory exercises.	Seoul, South Korea Mar 2025 – Jun 2025
Member & Mentor, Parrot Data Science Club SOGANG UNIVERSITY - Participated in machine learning and computer vision study groups, collaborating on data science projects. - Mentored five junior members in data analysis and CNN-based computer vision projects. - Organized Kaggle-style machine learning competitions and technical workshops for club members.	Seoul, South Korea Sep 2024 – Present
Teaching Assistant, English & Mathematics MEGASTUDY RUSSEL & YESUM ACADEMY Tutored middle and high school students while assisting classroom instruction in mathematics and English.	Seoul, South Korea Feb 2024 – Present
Satellite Operation Squad Leader REPUBLIC OF KOREA ARMY - Operated military satellite communication systems to ensure reliable communications across military units. - Led a team of seven personnel responsible for satellite equipment operation, maintenance, and inspection.	Daejeon, South Korea Jan 2022 – Jul 2023

Activities

Autonomous Taxi Design Project SOGANG UNIVERSITY - Designed an autonomous taxi system with emphasis on AI-based navigation, perception, and safety. - Proposed cost-effective sensor fusion strategies for reliable environmental perception.	Seoul, South Korea Jul 2025 – Aug 2025
Runner-up, AI Data Analysis Runnerthon SOGANG UNIVERSITY - Achieved 2nd place among 15 participating teams. - Developed an XGBoost-based real estate price prediction model through feature engineering and ensemble learning.	Seoul, South Korea Jan 2025 – Feb 2025
Software and AI Competency Enhancement Training Program REPUBLIC OF KOREA ARMY - Completed an intensive software and AI training program covering programming and machine learning fundamentals. - Developed a machine learning-based weather forecasting model using real-world meteorological data.	South Korea Jan 2023 – May 2023
English Training Program (R-EIP) HANKUK UNIVERSITY OF FOREIGN STUDIES (HUFS) Completed an intensive academic English program focused on communication and presentation skills.	Seoul, South Korea Oct 2021 – Dec 2021

Technical Skills

- Programming** Python, C/C++, SQL, R
- Machine Learning** PyTorch, TensorFlow, scikit-learn, Hugging Face
- Data & Computer Vision** NumPy, Pandas, OpenCV, Matplotlib
- Backend & Database** FastAPI, MySQL, ChromaDB
- Tools** Git, Linux, Docker, VS Code, Jupyter Notebook, Google Colab

Languages

- Korean:** Native
- English:** Fluent (TOEFL iBT 100, Feb 2026)
- Japanese:** Basic